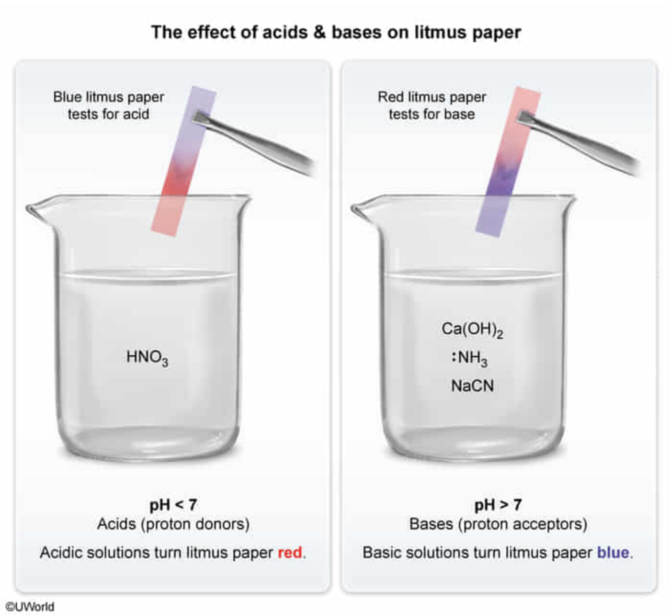
Litmus paper will turn blue when immersed in all of the following EXCEPT:

- A. $Ca(OH)_2$ [7%]
- B. NH_3 [8%]
- ✔ C. HNO_3 [71%]
- D. NaCN [12%]

Correct
71% answered correctly

Explanation:



Litmus paper is used as a pH **indicator** to determine whether a solution is acidic or basic. The paper is coated with litmus dye that changes color depending on the pH of the solution in which it is immersed. An acidic (pH < 7) solution will turn litmus paper red whereas a basic (pH > 7) solution will turn **litmus paper blue**.

The Brønsted-Lowry definitions of acids and bases can be used to identify acidic and basic molecules. Brønsted-Lowry acids donate a proton (H^+ ion), and Brønsted-Lowry bases accept a proton. The negatively charged ions OH^- and CN^- in $Ca(OH)_2$ and NaCN, respectively, can accept a proton **(Choices A and D)**. The N atom in NH_3 **(Choice B)** has a lone pair of electrons, which can accept an H^+ ion. Because these molecules are bases (proton acceptors), they will turn litmus paper blue.

HNO_3 can donate a proton, making it an acid. Therefore, when litmus paper is immersed in HNO_3, it will turn red rather than blue.

Educational objective:

Litmus paper is used as an acid-base indicator and changes color depending on the pH of the solution it is in. Brønsted-Lowry acids (pH < 7) can donate a proton (H^+ ion) and will turn litmus paper red. Brønsted-Lowry bases (pH > 7) can accept a proton and will turn litmus paper blue.

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